PV102 Day 8: Commissioning Troubleshooting and Professional Handoff

Students close PV102 by verifying a system from evidence: pre-energization inspection, polarity and continuity checks, voltage checks, functional checks, fault response, documentation, and client handoff.

PV102 | CTS Day 24 | Instructor Run Sheet
Test, document, and hand off what the system actually does. | Thursday, August 13, 2026

Core rule

Test, document, and hand off what the system actually does.

Learning targets

- Use a commissioning sequence to verify before energizing.
- Interpret basic test results without guessing beyond evidence.
- Troubleshoot from symptom to boundary to measurement.
- Prepare a concise handoff that names what passed, what failed, what changed, and what remains unknown.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Training kits or paper fault cards Multimeters Commissioning checklist Handoff form Labels Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 7 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.

12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work triggers.
1:00-2:15	Commissioning and Fault Handoff Challenge	Teams inspect a training system or fault card set, run a commissioning sequence, identify one fault, and deliver a final technical handoff.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 is complete. Carry the habit forward: define the job, verify the path, protect the boundary, document the truth.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use commissioning troubleshooting and professional handoff to make the next design decision safer and more defensible. The rule is simple: Test, document, and hand off what the system actually does."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Review the system scope and safety boundary.
- Complete pre-energization inspection using the checklist.
- Run only authorized de-energized checks first.
- Use evidence to identify the fault or missing documentation.
- Record pass/fail results and corrective action recommendation.
- Deliver a three-minute final handoff: system job, path, checks, fault, fix/next action, unknowns.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 is complete. Carry the habit forward: define the job, verify the path, protect the boundary, document the truth.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.